

2.2 Text Extraction and Cleanup

Pull words out of HTML and PDF, then fix encoding and boilerplate.

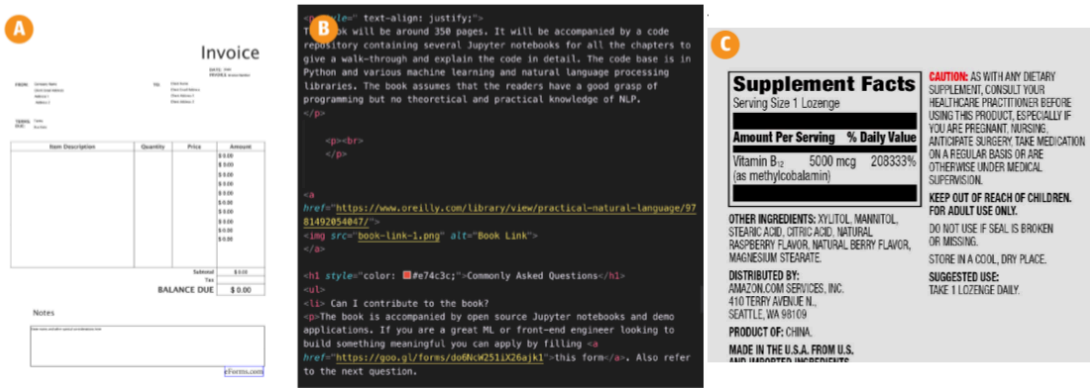
These notes walk through the lecture slides. The original deck is unchanged; use (slides) on the course hub if you want that PDF.

1. Pull the words out first

Acquisition gives you a file, a URL, or a scan. That is not yet text a tokenizer can use.

Text extraction is taking the human language out and throwing away the rest: tags, headers, page numbers, nav bars, image bytes. **Cleanup** is making that string usable: encoding, Unicode, spelling, format-specific junk.

This step is slow and it is application-dependent. A bad extract poisons every later box in the pipeline. Lab 2 is this note in code.



2. HTML: parse, do not regex the whole page

A typical page is a tree. The words you want sit in a few nodes. On a Stack Overflow question page, the question and the answers have their own tags and classes. Use those, then drop scripts, ads, and chrome.

Do not write a general HTML parser. Use **Beautiful Soup** or **Scrapy**. They give you the tree, CSS selectors, and encoding help.

What you keep depends on the job:

Keep	Drop (usually)
Title, question, accepted answer	Nav, footer, related-links sidebar
Article body	Cookie banners, share widgets
User comment text	Tracking pixels, JSON-LD you do not need

Boilerplate is the repeated template around the article. If you leave it in, "Home", "Subscribe", and "Cookie policy" become high-count tokens and the model learns the site chrome, not the document.

3. Encoding and Unicode

Files lie about their encoding. `open(path)` with the wrong encoding turns `é` into `Ã©` or into a crash. Default to **UTF-8**. If a file is old Windows text, try `cp1252` or `latin-1` only after UTF-8 fails. In Python:

```
with open("page.html", "r", encoding="utf-8") as f:
    raw = f.read()
```

Unicode normalization folds characters that look the same but are stored differently (`é` as one code point vs `e` + combining accent). Emoji and other symbols are Unicode too. If you strip anything outside ASCII you delete meaning: a green heart, a warning sign, a name in another script.

 U+2191	 U+1F647	 U+2010	 U+FF64	 U+0398	 U+1F49A	 U+263B	 U+056E	 U+0C2C	 U+0CA0
 U+03B3	 U+12CE	 U+1F49C	 U+03BC	 U+1F680	 U+266A	 U+FE36	 U+30FB	 U+10E6	 U+2036
 U+263C	 U+0964	 U+26AC	 U+0939	 U+4E14	 U+09F0	 U+0F4F	 U+2030	 U+30C7	 U+21A9
 U+2018	 U+2033	 U+026A	 U+0DA2	 U+1F639	 U+0394	 U+00F9	 U+27AB	 U+0177	 U+1F9D8

A practical rule: decode to Unicode once, normalize (NFC is the usual choice), then decide what to keep. Do not `.encode("ascii", "ignore")` unless you have a reason.

4. PDF and scans

Digital PDFs (exported from Word or LaTeX) have a text layer. `pypdf` or `PDFMiner` can pull it. Columns, headers, and footnotes often come out in the wrong order. Hyphenated line breaks (`process-` / `ing`) need a join rule.

Scanned PDFs are pictures. You need OCR (optical character recognition). Expect errors: `rn` for `m`, lost tables, garbage near logos.

Libraries are far from perfect. Some PDFs will not parse. That is a data problem, not a "you used the wrong import" problem. Print a page of raw extract and read it before you tokenize.

5. Spelling and system-specific junk

Spell correction is optional and it is not reliable. A dictionary checker will "fix" names and domain terms. If you build one, train it on *this* language and *this* domain, then spot-check.

Other cleanup that shows up in real feeds:

- HTML entities (`&` → `&`)
- soft hyphens and zero-width characters
- OCR letter swaps
- repeated whitespace and leftover markup (`<br>` , markdown hashes)

Do the smallest cleanup that makes the next stage honest. Over-cleaning deletes signal (emoji in sentiment, `#hashtag` in tweets).

6. Practice

1. You extract a news article with BeautifulSoup and the word `Subscribe` is among the top ten tokens. What did you leave in?
2. Why is `text.encode("ascii", "ignore").decode("ascii")` a bad default for tweets?
3. A PDF of a two-column paper comes out as line 1 of the left column, line 1 of the right column, line 2 of the left column, ... What went wrong, and what do you check before you train?